

Alguns Artigos da Área de Energia Com Aplicações de Identificação de Sistemas

1 - Federico Cassola a, Massimiliano Burlando , “Wind speed and wind energy forecast through Kalman filtering of Numerical”, Applied Energy, 154-166 (2012).

Despite the major progress made by Numerical Weather Prediction (NWP) in the last decades, meteorological models are usually unable to provide reliable surface wind speed forecasts, especially in complex topography regions, because of shortcomings in horizontal resolution, physical parameterisations, initial and boundary conditions. In order to reduce these drawbacks, one of the most successful approaches is the Kalman filtering technique, which combines recursively observations and model forecasts to minimise the corresponding biases. In meteorology, Kalman filters are widely used to improve the prediction of variables characterised by well-defined cyclicities, whereas the evolution of wind speed is usually too irregular. In the present paper, the Kalman filter is analysed in order to find the best configuration for wind speed and wind power forecast. The procedure has been tested, in a hindcast mode, with 2-yearlong data sets of wind speed provided by a NWP model and two anemometric stations located in the eastern Liguria (Italy). It is shown that, tuning time step and forecast horizon of the filter, this methodology is capable to provide significant forecast improvement with respect to the wind speed model direct output, especially when used for very short-term forecast. In this configuration, Kalman-filtered wind speed data have been used to forecast the wind energy output of the nearby wind farm of Varese Ligure. After 2 years of testing, the percentage error between simulated and measured wind energy values was still very low and showed a stable evolution.

2 - Hui Liu, Hong-qi Tian, Yan-fei Li, “Comparison of two new ARIMA-ANN and ARIMA-Kalman hybrid methods for wind speed prediction,” Applied Energy, 415-424 (2012).

Wind speed prediction is important to protect the security of wind power integration. The performance of hybrid methods is always better than that of single ones in wind speed prediction. Based on Time Series, Artificial Neural Networks (ANN) and Kalman Filter (KF), in the study two hybrid methods are proposed and their performance is compared. In hybrid ARIMA-ANN model, the ARIMA model is utilized to decide the structure of an ANN model. In hybrid ARIMA-Kalman model, the ARIMA model is employed to initialize the Kalman Measurement and the state equations for a Kalman model. Two cases show both of them have good performance, which can be applied to the non-stationary wind speed prediction in wind power systems.

3 - Byung O Kang, Jae H. Park, “Kalman Filter MPPT Method for a Solar Inverter,” Power and Energy Conference at Illinois (PECI), 2011 IEEE, (2011)

This paper proposes a new maximum power point tracking (MPPT) method for photovoltaic (PV) systems by using Kalman filter. A Perturbation & Observation (P&O) method is widely used presently due to its easy implementation and simplicity. The P&O usually requires of dithering scheme to reduce noise effects, but it slows the tracking response. Tracking speed is the most important factor on improving efficiency in frequent environmental changes. The proposed method is based on the Kalman filter. It shows the fast tracking performance on noisy conditions, so that enables to generate more power in rapid weather changes than the P&O. Simulation results are provided the comparison between the proposed method and P&O on time responses for conditions of sudden system restart and sudden irradiance change.

4 - Varun Ramchandania, Kranthi Pamarthib, Naveen Varma, Shubhajit Roy Chowdhury, “Implementation of Maximum Power Point Tracking Using Kalman Filter for Solar Photovoltaic Array on FPGA”, International Journal of Smart Grid and Clean Energy , vol. 2, no. 2, May 2013: pp. 152–158.

This paper proposes FPGA implementation of a novel approach to track maximum power point of a solar photovoltaic array. The approach uses Kalman filter algorithm to track maximum power point. Using this approach tracking becomes much faster than using the generic Perturb & Observe algorithm in case of sudden weather changes. In this paper output of the proposed algorithm on FPGA is provided. Experimentation was performed under optimal conditions as well as under cloudy conditions i.e. falling irradiance levels. Using the proposed technique the maximum power point of a solar PV array is tracked with an efficiency of 97.11%. Moreover, the maximum power point has been tracked at a much faster rate i.e. 4.5 ms using the proposed algorithm compared to the existing generic Perturb and Observe approach.

5 - Abdelouahed Mesbahi, Abdelhadi Raihani, Omar Bouattane, Abdellah Saad, Mohamed Khafallah, “Extended Kalman Filter for characterizing a Wind Energy Conversion System based on Variable Speed Permanent Magnet Synchronous Generator”, 2014 International Renewable and Sustainable Energy Conference (IRSEC)

This paper presents an approach to determine the power coefficient for a standalone wind energy conversion system (WECS) and estimate the wind speed. The knowledge of the power coefficient requires absolutely estimating the mechanical power supplied by the wind turbine of the system. An Extended Kalman Filter (EKF) associated to a PMSG model is used to reconstruct angular velocity and position. Generally the WECS manufacturer provides the power output characteristic. Hence, the wind speed and power coefficient can be deduced by knowing the rotor angular speed. All the used measurement points are stored as a database to allow a deep study and a good estimation and knowledge of the system.

6 - Abhisek Ukil , Rastko Zivanovi', "The detection of abrupt changes using recursive identification for power system fault analysis", *Electric Power Systems Research* 77 259–265 (2007).

This paper describes the application of the recursive parameter estimation technique used to detect the abrupt changes in the signals recorded during disturbances in the power network of South Africa. The recursive identification technique uses M parallel Kalman filters. Main focus has been to estimate the time-instants of the changes in the signal model parameters during the pre-fault condition and following the events like initiation of fault, circuit-breaker opening, auto-reclosure of the circuit-breakers and the like. After segmenting the fault signal precisely into these event-specific sections, further signal processing and analysis can be performed on these segments, leading to automated fault recognition and analysis. In the scope of this paper, we focus on the first task, that is, segmenting the fault signal into event-specific sections using the recursive identification technique.

7 - Maamar Bettayeb , Uvais Qidwai, "Recursive estimation of power system harmonics", *Electric Power Systems Research* 47, 143–152 (1998).

This paper presents the application of well known recursive estimation techniques to the important problem of power system harmonics in a noisy environment. on-line estimation of harmonic amplitudes and phases is performed using several variants of recursive least square (RLS) algorithms, known for their simplicity of computation and good convergence properties. The estimates are updated recursively as samples of the harmonic signals are received. A noisy harmonic signal from the AC bus of a six pulse rectifier is used as a test signal in the simulation. The various RLS algorithms are evaluated under different signal to noise ratios (SNR) and are shown to produce good harmonic magnitude and phase estimates even for a 0 dB SNR. Due to their simplicity, these algorithms are appropriate for on-line implementation in polluted power systems.

8 - Haili Ma, Adly A. Girgis, "IDENTIFICATION AND TRACKING OF HARMONIC SOURCES IN A POWER SYSTEM USING A KALMAN FILTER", *IEEE Transactions on Power Delivery*, Vol. 11, No. 3, July 1996.

In this paper, two problems have been addressed on harmonic sources identification: the optimal locations of a limited number of harmonic meters and the optimal dynamic estimates of harmonic source locations and their injections in unbalanced three-phase power systems. A Kalman filtering is used to attack these problems. System error covariance analysis by the Kalman filter associated with a harmonic injection estimate determines the optimal arrangement of limited harmonic meters. Based on the optimally-arranged harmonic metering locations, the Kalman filter then yields the optimal dynamic estimates of harmonic injections with a few noisy harmonic measurements. The method is dynamic and has the capability of identifying, analyzing and tracking each harmonic injection at all buses in unbalanced three-phase power systems. Actual recorded harmonic measurements and simulated data in a power distribution system are provided to prove the efficiency of this approach.

9 - Weidong Xiao, Magnus G. J. Lind, William G. Dunford, "Real-Time Identification of Optimal Operating Points in Photovoltaic Power Systems", *IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS*, VOL. 53, NO. 4, AUGUST 2006.

Photovoltaic power systems are usually integrated with some specific control algorithms to deliver the maximum possible power. Several maximum power point tracking (MPPT) methods that force the operating point to oscillate have been presented in the past few decades. In the MPPT system, the ideal operation is to determine the maximum power point (MPP) of the photovoltaic (PV) array directly rather than to track it by using the active operation of trial and error, which causes undesirable oscillation around the MPP. Since the output features of a PV cell vary with environment changes in irradiance and temperature from time to time, real-time operation is required to trace the variations of local MPPs in PV power systems. The method of real-time estimation proposed in this paper uses polynomials to demonstrate the power–voltage relationship of PV panels and implements the recursive least-squares method and Newton–Raphson method to identify the voltage of the optimal operating point. The effectiveness of the proposed methods is successfully demonstrated by computer simulations and experimental evaluations of two major types of PV panels, namely: 1) crystalline silicon and 2) copper–indium–diselenide thin film.